PRIMER IZPITA PRI PREDMETU PROGRAMIRANJE / PROGRAMIRANJE 2

1. (10 točk) Zakaj so drevesa uporabna za iskanje podatkov?
2. (10 točk) Kaj je graf (v okviru informacijske tehnologije seveda!)? Zakaj so grafi uporabni?
3. (20 točk) Napiši `void sortDescending(std::vector<int>& v)`, ki uredi padajoče (brez std::sort seveda!)
4. (20 točk) Napiši `void insertAt(std::vector<int>& v, int index, int value)`, ki vstavi element na indeks (če je indeks veljaven ali na konec)
5. (20 točk) Narišite/Zapišite korake kako algoritem 'Merge sort' uredi sledeč vektor { -3, 4, 5, -6, 0, 8 }
6. (20 točk) Narišite in opišite kako se v binarno binarno drevo dodajo sledeči elementi { -3, 4, 5, -6, 0, 8 }.

SAMPLE EXAM FOR THE COURSE PROGRAMMING / PROGRAMMING 2

1. (10 points) Why are trees useful for searching data?
2. (10 points) What is a graph, in the context of information technology? Why are graphs useful?
3. (20 points) Write `void sortDescending(std::vector<int>& v)`, which sorts the vector in descending order, without using std::sort, of course.
4. (20 points) Write `void insertAt(std::vector<int>& v, int index, int value)`, which inserts an element at the given index, if the index is valid, or at the end otherwise.
5. (20 points) Draw/write the steps showing how the Merge sort algorithm sorts the following vector: { -3, 4, 5, -6, 0, 8 }
6. (20 points) Draw and describe how the following elements are added to a binary tree:{ -3, 4, 5, -6, 0, 8 }